

PART - B

JAVA LAB PROGRAMS

1. A Cloth showroom has announced the following seasonal discount on purchase of items.

Purchase Amount	Discount	
	Mill Cloth	Handloom items
0 - 100	-	5.0%
101 - 200	5.0%	7.5%
201 - 300	7.5%	10.0%
Above 300	10.0%	15.0%

Write a program using switch and if statements to compute the net amount to be paid by a customer.

```
import java.io.*;
class Showroom
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("1. For Mill Cloth");
        System.out.println("2. For Handloom items");

        System.out.println("Enter your choice...");
        int ch = Integer.parseInt(br.readLine());

        System.out.println("Enter the purchase amount...");
        int amount = Integer.parseInt(br.readLine());

        double discount = 0, netAmount = 0;

        switch(ch)
        {
            case 1:
                if (amount <= 100)
```

```

        discount = 0;
    if (amount >=101 && amount <=200)
        discount = amount * 5.0 / 100;
    if (amount >=201 && amount <=300)
        discount = amount * 7.5 / 100;
    if (amount > 300)
        discount = amount * 10.0 / 100;

    netAmount = amount - discount;
    System.out.println("Purchase Amount: " + amount);
    System.out.println("Discount: " + discount);
    System.out.println("Amount paid by the customer: " +
netAmount);

    break;

case 2:
    if (amount <= 100)
        discount = amount * 5.0 / 100;
    if (amount >=101 && amount <=200)
        discount = amount * 7.5 / 100;
    if (amount >=201 && amount <=300)
        discount = amount * 10.0 / 100;
    if (amount > 300)
        discount = amount * 15.0 / 100;

    netAmount = amount - discount;
    System.out.println("Purchase Amount: " + amount);
    System.out.println("Discount: " + discount);
    System.out.println("Amount paid by the customer: " +
netAmount);

    break;

default:
    System.out.println("Invalid choice");
}
}
}

```

Output:

```

javac Showroom.java
java Showroom

```

1. For Mill Cloth
2. For Handloom items

Enter your choice...

2

Enter the purchase amount...

25000

Purchase Amount: 25000

Discount: 3750.0

Amount paid by the customer: 21250.0

java Showroom

1. For Mill Cloth

2. For Handloom items

Enter your choice...

1

Enter the purchase amount...

15000

Purchase Amount: 15000

Discount: 1500.0

Amount paid by the customer: 13500.0

2. Write a program to find first N values of Fibonacci Series.

```
import java.io.*;
class Fibonacci
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        System.out.println("Enter the N value for Fibonacci series...");
        int n = Integer.parseInt(br.readLine());

        int a = 0, b = 1, c;
        System.out.println("Fibonacci series are:");
        System.out.print(a + " " + b + " ");

        int i = 3;
        do
        {
            c = a + b;
            System.out.print(c + " ");
            a = b;
```

```

        b = c;
        i++;
    }
    while(i <= n);
}
}

```

Output:

```

javac Fibonacci.java
java Fibonacci

```

```

Enter the N value for Fibonacci series...
10
Fibonacci series are:
0 1 1 2 3 5 8 13 21 34

```

3. Write a program to accept A and B matrices and produce the third matrix $C = A \times B$. (Multiplication or Product of Two Matrix)

```

import java.io.*;
class Product
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        int a[][] = new int[10][10];
        int b[][] = new int[10][10];
        int c[][] = new int[10][10];

        System.out.println("Enter the number of rows and columns for first
matrix:");
        int r1 = Integer.parseInt(br.readLine());
        int c1 = Integer.parseInt(br.readLine());

        System.out.println("Enter the number of rows and columns for second
matrix:");
        int r2 = Integer.parseInt(br.readLine());
        int c2 = Integer.parseInt(br.readLine());
    }
}

```

```

if(c1 != r2)
{
    System.out.println("Matrix multiplication is not possible.");
    System.exit(0);
}

System.out.println("Enter the elements of first matrix:");
for(int i = 0; i < r1; i++)
{
    for(int j = 0; j < c1; j++)
    {
        a[i][j] = Integer.parseInt(br.readLine());
    }
}

System.out.println("Enter the elements of second matrix:");
for(int i = 0; i < r2; i++)
{
    for(int j = 0; j < c2; j++)
    {
        b[i][j] = Integer.parseInt(br.readLine());
    }
}

// Perform matrix multiplication
for(int i = 0; i < r1; i++)
{
    for(int j = 0; j < c2; j++)
    {
        c[i][j] = 0;
        for(int k = 0; k < c1; k++)
        {
            c[i][j] += a[i][k] * b[k][j];
        }
    }
}

// Display the result
System.out.println("Resultant matrix:");
for(int i = 0; i < r1; i++)
{
    for(int j = 0; j < c2; j++)
    {
        System.out.print(c[i][j] + " ");
    }
}

```

```
        System.out.println();
    }
}
}
```

Output:

```
javac Product.java
java Product
```

Enter the number of rows and columns for first matrix:

2

3

Enter the number of rows and columns for second matrix:

3

2

Enter the elements of first matrix:

3

3

3

3

3

3

Enter the elements of second matrix:

4

4

4

4

4

Resultant matrix:

36 36

36 36

```
java Product
```

Enter the number of rows and columns for first matrix:

3

2

Enter the number of rows and columns for second matrix:

4

2

Matrix multiplication is not possible.

**4. Write a program to accept N String. Sort them in Ascending Order.
(Alphabetical Order)**

```
import java.io.*;
class Alphabetical
{
    public static void main(String args[]) throws IOException
    {
        InputStreamReader isr = new InputStreamReader(System.in);
        BufferedReader br = new BufferedReader(isr);

        String str[] = new String[10];
        System.out.println("Enter total number of strings:");
        int n = Integer.parseInt(br.readLine());

        System.out.println("Enter the strings:");
        for(int i = 0; i < n; i++)
        {
            str[i] = br.readLine();
        }

        // Sort the strings in alphabetical order
        for(int i = 0; i < n - 1; i++)
        {
            for(int j = i + 1; j < n; j++)
            {
                if(str[i].compareTo(str[j]) > 0)
                {
                    String temp = str[i];
                    str[i] = str[j];
                    str[j] = temp;
                }
            }
        }

        System.out.println("Strings in alphabetical order:");
        for(int i = 0; i < n; i++)
        {
            System.out.println(str[i]);
        }
    }
}
```

Output:

```
javac Alphabetical.java
java Alphabetical
```

Enter total number of strings:

5

Enter the strings:

hp

sony

lenovo

acer

dell

Strings in alphabetical order:

acer

dell

hp

lenovo

sony

5. Write a program to demonstrate usage of the classes and objects.

```
class Calculate
{
    int length, breadth;
    void getData(int x, int y)
    {
        length = x;
        breadth = y;
    }
    int calcArea()
    {
        int area = length * breadth;
        return area;
    }
}

class Rectangle
{
    public static void main(String args[])
    {
        Calculate c1 = new Calculate();
        c1.getData(10, 20);
        int area = c1.calcArea();
    }
}
```

```

        System.out.println("Area of rectangle: " + area);

        Calculate c2 = new Calculate();
        c2.length = 5;
        c2.breadth = 10;
        int area2 = c2.length * c2.breadth;
        System.out.println("Area of rectangle: " + area2);
    }
}

```

Output:

```

javac Rectangle.java
java Rectangle

```

```

Area of rectangle: 200
Area of rectangle: 50

```

6. Write a program to implement the concepts of multiple inheritance.

```

class Student
{
    int rollno;
    void getNumber(int n)
    {
        rollno = n;
    }
    void displayNumber()
    {
        System.out.println("Roll Number = " + rollno);
    }
}

class Test extends Student
{
    float part1, part2;
    void getMarks(float m1, float m2)
    {
        part1 = m1;
        part2 = m2;
    }
    void displayMarks()

```

```

    {
        System.out.println("Marks in Part 1 = " + part1);
        System.out.println("Marks in Part 2 = " + part2);
    }
}

interface Sports
{
    float sportwt = 6.0f;
    void displaySportswt();
}

class Result extends Test implements Sports
{
    float total;
    public void displaySportswt()
    {
        System.out.println("Sports Weightage = " + sportwt);
    }
    void displayTotal()
    {
        total = part1 + part2 + sportwt;
        displayNumber();
        displayMarks();
        displaySportswt();
        System.out.println("Total Score = " + total);
    }
}

class Multiple
{
    public static void main(String args[])
    {
        Result student1 = new Result();
        student1.getNumber(101);
        student1.getMarks(85.5f, 90.0f);
        student1.displayTotal();
    }
}

```

Output:

```

javac Multiple.java
java Multiple

```

Roll Number = 101

Marks in Part 1 = 85.5
Marks in Part 2 = 90.0
Sports Weightage = 6.0
Total Score = 181.5

7. Write a program to demonstrate the concepts of multithreaded programming.

```
class A extends Thread
{
    public void run()
    {
        for (int i = 1; i <=5; i++)
        {
            if (i == 1)
                yield();
            System.out.println("\tFrom Thread A : i = " + i);
        }
        System.out.println("Exit from Thread A");
    }
}

class B extends Thread
{
    public void run()
    {
        for (int j = 1; j <=5; j++)
        {
            System.out.println("\tFrom Thread B : j = " + j);
            if (j == 3)
                stop();
        }
        System.out.println("Exit from Thread B");
    }
}

class C extends Thread
{
    public void run()
    {
        for (int k = 1; k <=5; k++)
        {
```

```

        System.out.println("\tFrom Thread C : k = " + k);
        if (k == 1)
            try
            {
                sleep(1000);
            }
            catch (Exception e)
            {
                System.out.println("Thread C interrupted");
            }
        System.out.println("Exit from Thread C");
    }
}

class Multithread
{
    public static void main(String args[])
    {
        A threadA = new A();
        B threadB = new B();
        C threadC = new C();

        System.out.println("Starting Thread A");
        threadA.start();
        System.out.println("Starting Thread B");
        threadB.start();
        System.out.println("Starting Thread C");
        threadC.start();
        System.out.println("End of main thread");
    }
}

```

Output:

```

javac Multithread.java
java Multithread

```

```

Starting Thread A
Starting Thread B
    From Thread A : i = 1
Starting Thread C
    From Thread B : j = 1
    From Thread A : i = 2
End of main thread
    From Thread B : j = 2

```

```
From Thread C : k = 1
From Thread A : i = 3
From Thread B : j = 3
From Thread A : i = 4
From Thread A : i = 5
Exit from Thread A
From Thread C : k = 2
From Thread C : k = 3
From Thread C : k = 4
From Thread C : k = 5
Exit from Thread C
```

8. Write an applet program to find the sum of two integers using user input.

Userinput.java

```
import java.awt.*;
import java.applet.*;

public class Userinput extends Applet
{
    TextField text1, text2;

    public void init()
    {
        text1 = new TextField(10);
        text2 = new TextField(10);
        add(text1);
        add(text2);
    }

    public void paint(Graphics g)
    {
        int x = 0, y = 0, z = 0;
        String s1, s2, s;

        try
        {
            g.drawString("Input a number in each box and press enter key",
```

```

10, 50);

        s1 = text1.getText();
        x = Integer.parseInt(s1);
        s2 = text2.getText();
        y = Integer.parseInt(s2);
    }
    catch(Exception e)
    {
    }

    z = x + y;
    s = String.valueOf(z);
    g.drawString("The Sum is :", 10, 75);
    g.drawString(s, 100, 75);
}

public boolean action(Event e, Object ob)
{
    repaint();
    return (true);
}
}

```

Userinput.html

```

<Html>
<Head>
    <Title>User Input</Title>
</Head>
<Body>
<Center>
    <Applet code=Userinput.class Width=400 Height=400>
    </Applet>
</Center>
</Body>
</Html>

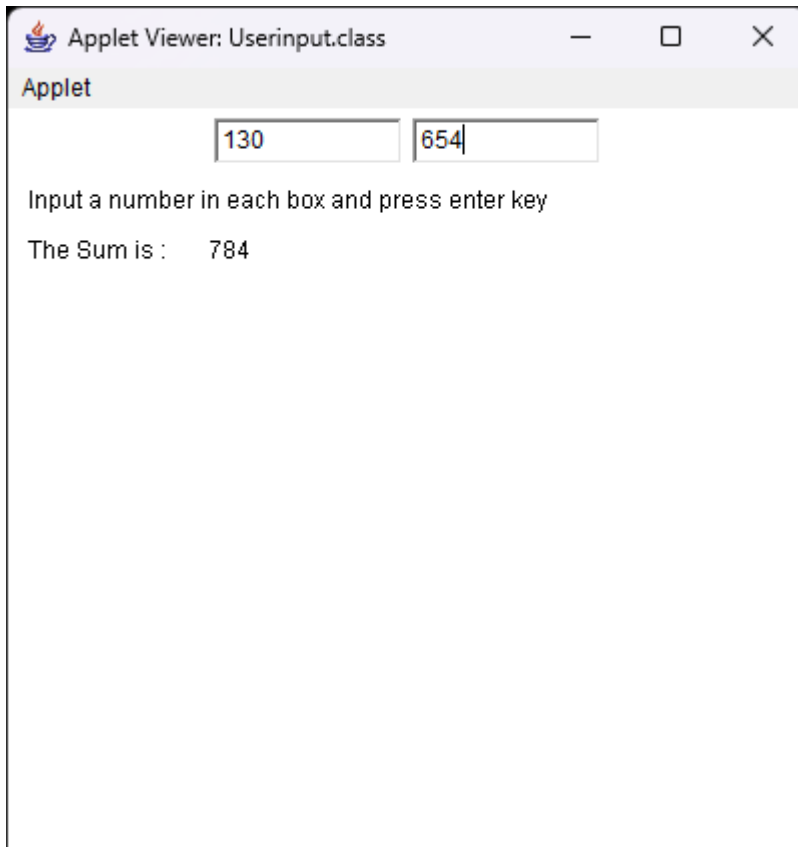
```

Output:

```

javac Userinput.java
appletviewer Userinput

```



9. Write an applet program to draw a human face.

Face.java

```
import java.awt.*;
import java.applet.*;

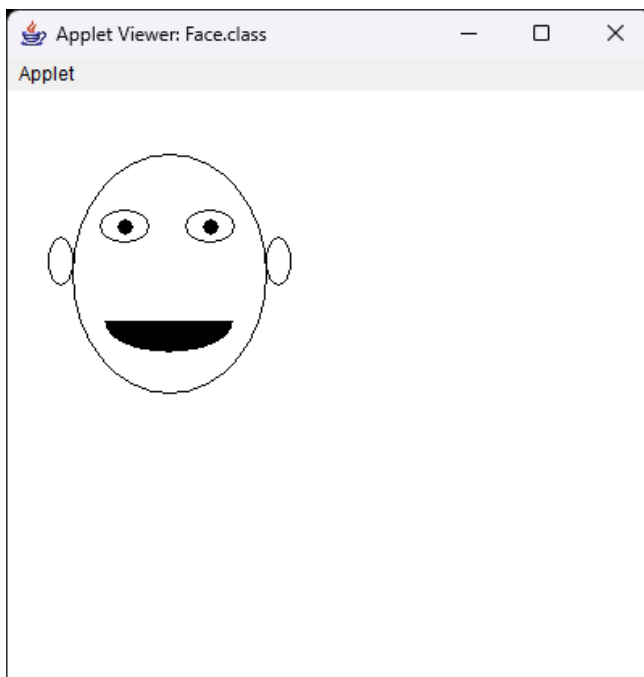
public class Face extends Applet
{
    public void paint (Graphics g)
    {
        g.drawOval(40,40,120,150);
        g.drawOval(57,75,30,20);
        g.drawOval(110,75,30,20);
        g.fillOval(68,81,10,10);
        g.fillOval(121,81,10,10);
        g.fillArc(60,125,80,40,180,180);
        g.drawOval(25,92,15,30);
        g.drawOval(160,92,15,30);
    }
}
```

Face.html

```
<Html>
<Head>
  <Title>Human Face</Title>
</Head>
<Body>
<Center>
  <Applet code=Face.class Width=400 Height=400>
  </Applet>
</Center>
</Body>
</Html>
```

Output:

```
javac Face.java
appletviewer Face
```



10. Write an applet program to create a Bar chart using the following table.

Year	2012	2013	2014	2015
Result Analysis (Maximum 100%)	90	95	93	100

ExBarchart.java

```
import java.awt.*;
import java.applet.*;

public class ExBarchart extends Applet
{
    int n = 0;
    String label[];
    int value[];

    public void init()
    {
        try
        {
            n = Integer.parseInt(getParameter("columns"));
            label = new String[n];
            value = new int[n];

            label[0] = getParameter("label1");
            label[1] = getParameter("label2");
            label[2] = getParameter("label3");
            label[3] = getParameter("label4");

            value[0] = Integer.parseInt(getParameter("c1"));
            value[1] = Integer.parseInt(getParameter("c2"));
            value[2] = Integer.parseInt(getParameter("c3"));
            value[3] = Integer.parseInt(getParameter("c4"));
        }
        catch(NumberFormatException e)
        {
        }
    }

    public void paint(Graphics g)
    {
        for(int i = 0; i < n; i++)
        {
            g.setColor(Color.blue);
            g.drawString(label[i], 20, i * 50 + 30);
            g.fillRect(50, i * 50 + 10, value[i], 40);
        }
    }
}
```

ExBarchart.html

```
<Html>
<Head>
  <Title> Drawing Bar Chart </Title>
</Head>
<Body>
<Center>
<applet code="ExBarchart.class" width=500 height=500>
<PARAM NAME = "columns" VALUE = "4">
<PARAM NAME = "c1" VALUE = "90">
<PARAM NAME = "c2" VALUE = "95">
<PARAM NAME = "c3" VALUE = "93">
<PARAM NAME = "c4" VALUE = "100">
<PARAM NAME = "label1" VALUE = "2012">
<PARAM NAME = "label2" VALUE = "2013">
<PARAM NAME = "label3" VALUE = "2014">
<PARAM NAME = "label4" VALUE = "2015">
</applet>
</Center>
</Body>
</Html>
```

Output:

```
javac ExBarchart.java
appletviewer ExBarchart.html
```

